Birla Institute of Technology and Science, Pilani
1st Semester 2025-2026
CS F111 – Computer Programming – End-semester (Closed Book, Part-A)

13 Dec 2025 | Weightage: 35% | Duration: 3 hours | Max. marks: 76

Student Name: [redacted] ID No.: [redacted] Room no. [redacted]

Student Sign.: [redacted] Invigilator Sign.: [signature] Marks obtained: [redacted] /76

Q.1		Q.2	Q.3	Q.4	Re-check request (if any)
2 x	-0.5 x				
[redacted]					

[**Instructions:** There are total 4 questions. **Assume that #include<stdio.h> is included at the top of each program.** Please WRITE YOUR ANSWERS **CLEARLY IN THE SPACE PROVIDED.** Answers written at any other place will not be evaluated. Use the rough sheet provided for rough work. **Note: For MCQs only 1 answer is correct,** and there is **25 % negative marking for each wrong answer.** If more than 1 answer is written, then that question will not be evaluated.]

[2 marks questions] [For Q. 1, i - v, -0.5 MARKS FOR EACH WRONG ANSWER] Please tick mark the answer properly. ☑ **[10x2=20]**

Q.1 Multiple choice questions:

i). Convert $(524)_8$ to base 8.
a. $(264)_8$ ☐ b. $(304)_8$ ☒ NA c. $(312)_8$ ☐ d. $(244)_8$ ☐

ii). Convert $(A3FD)_{16}$ to decimal.
a. 40981 ☐ b. 41981 ☒ NA c. 42981 ☐ d. 41891 ☐

iii). What is the output of the following code?
```
int main()
{   int arr[] = {10, 20, 30, 40};
int i = 0;
printf("%d ", arr[i++]);
printf("%d ", arr[++i]);
printf("%d", arr[i++]);
return 0;
}
```
a. 10 30 30 ☑ b. 10 20 30 ☐ c. 10 30 40 ☐ d. 20 30 40 ☐

iv). What is the output of the following code?
```
int main()
{
int A = 10;
A = A + 'B';
printf("%d %d", A, 'A');
return 0;
}
```
a. 66 65 ☐ b. 10 65 ☐ c. 77 66 ☐ d. 76 65 ☑

v). The value of the 32-bit IEEE 754 FPR 0 10001000 11110101000000000011000 in hexadecimal is____.
a. 0x447A8000 ☐ b. 0x44790018 ☐ c. 0x447A8018 ☒ NA d. 0x447B0018 ☐

vi). How many times will "hello" be printed?
```
int main()
{
int i, j, k;
for(i = 1; i <= 4; i++)
    for(j = 1; j <= i; j++)
        for(k = 1; k <= j; k++)
            printf("hello\n");
return 0;
}
```
a. 10 ☐ b. 15 ☐ c. 20 ☑ d. 24 ☐

vii). What is the output of the code?
```
int main()
{
int i, j;
for(i = 1; i <= 5; i++) {
    if(i == 3) continue;
    for(j = 1; j <= i; j++) {
        if(j == 2) break;
        printf("*");
    }
}

return 0;
}
```
a. **** ☑ b. ***** ☐ c. *** ☐ d. ****** ☐

viii). What is the output of the code?

```
int main()
{
int i, j;
for(i = 1; i <= 4; i++) {
    for(j = 1; j <= 4 - i; j++)
        printf(" ");
    for(j = 1; j <= 2*i - 1; j++)
        printf("*");
    printf("\n");
}
return 0;}
```

a. Right-aligned triangle ☐ b. Centered pyramid ☑ c. Inverted triangle ☐ d. Inverted pyramid ☐

ix). What is the output of the code?

```
int main()
{
int a = 5, b = 3, c = 2;
int result = a > b ? (b > c ? b : c) : a;
printf("%d", result);
return 0;}
```

a. 5 ☐ b. 2 ☐ c. Compilation error ☐ d. 3 ☑

x). What is the output of the code?

```
#include<string.h>
int main() {
 char a[15];
 char *s = "Hello BITS";
 int len = strlen(s);
 for(int i = 0; i < len; i++)
    a[i] = s[len - 1 - i];
 a[len] = '\0';
 printf("%s", a);
return 0;}
```

a. Hello BITS ☐ b. olleH STIB ☐ c. BITS Hello ☐ d. STIB olleH ☑

[6 marks questions] [For Qs. 2, i-v there is no negative marking. Please note that the outputs are case-sensitive]

Q.2. i). Given the output of the following code is "1 2 4 5 7 9 10 6 ", fill in the blanks (___) [3x2=6]

```
#include<stdlib.h>

void mysort(int a[], int SIZE) {
  int temp, i , j, sorted = 0;
 for(i = 0; i < SIZE; i++){
  for(j = SIZE-1; j >i; j--){
      if(a[j-1] > a[j])
      {
a).     temp= a[j];
b).     a[j] = a[j-1];
c).     a[j-1]= temp;
      }}}}
```

```
int main() {
      int arr[8] = {2,5,9,7,1,10,4,6};
      int SIZE = 8;
      int i;
      mysort(arr, SIZE);
      for (i = 0; i < SIZE; i++)
          {printf("%d ", arr[i]);
           }
      return 0;
}
```

ii). Fill the blanks for the following code? Output: 40 50 20 40 40 [3x2=6]

```
int main() {
    int arr[] = {10, 20, 30, 40, 50};
    arr[2] = *(arr + 1);
    *(arr + 1) = ????????;
    int *p;
    p = &arr[5];
    ?????????

    ?????????
    *arr = *p;
    for (int i = 0; i < 5; i++)
        printf("%d ", *(arr + i));
    return 0;
}
```

//a). arr[4]

//b). p--;

//c). *p = *(arr+3);

[3x2=6]

iii). What is the output of the following code?

```
#define MAX(A, B) ((A > B) ? (A) : (B))
int main() {
    int max_val,x,y;
    x = 5;
    y = 10;
    max_val = MAX(x++, y++);
    printf("The result is: %d\n", max_val);
    printf("x is %d\n", x);
    printf("y is %d", y);
    return 0;}
```

//a). The result is: 10

//b). x is 6

//c). y is 11

iv). What is the output of the following code? [3x2=6]

```
int main()
{
    int arr[] = {10, 12, 14};
    int *p, **pp;
    p = arr;
    pp = &p;
    printf("%d, %d\n", *(p + 1), *(*pp + 2));
    printf("%d, %d\n", *p + 1, **pp + 2);
    printf("%d", (*pp)[1]);
    return 0;
}
```

//a). 12, 14 ✓ (2)
//b). _____
//c). N.A

v). Fill all the blanks (1 mark will be awarded for each row only if entire row is correct, otherwise 0 marks for that row. [6x1=6]

If File Handling Mode is:	Must the file exist? (Yes/No)	Can it create a file? (Yes/No)	Can it truncate a file? (Yes/No)	Read-ability (Yes/No)	Write-ability (Yes/No)	Append-ability (Yes/No)
"r"	Yes	No	No	Yes	No	No
"r+"	Yes	No	No	Yes	Yes	No
"w"	No	Yes	Yes	No	Yes	No
"w+"	No	Yes	Yes	Yes	Yes	No
"a"	No	Yes	No	No	Yes	Yes
"a+"	No	Yes	No	Yes	Yes	Yes

(6)

Q.3. Fill the blanks in the following program. [7x2 = 14]

Output:
```
Sorted Array '*b':
1    2    3    4    5
```

```
#include <stdlib.h>
void sortAndCopy(int c[], int n, int **d) {
    *d = (int *) malloc(????????);
    if (*d == NULL) {
        printf("Memory allocation failed!\n");
        return;
    }
    for (int i = 0; i < n; i++) {
        (*d)[i] = c[i];
    }
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            // Compare adjacent elements
            if (????????) {
                ?????? ;
                (*d)[j] = ???????;
                ????????;
            }
        }
    }
}

int main() {
    int a[5] = {1, 5, 4, 2, 3};
    int **b = (int **) malloc(sizeof(int *));
    sortAndCopy(a, 5, b);
    printf("Sorted Array '*b':\n");
    for(int i = 0; i < 5; i++) {
        printf("%d\t", (*b)[i]);
    }
    printf("\n");
    free(??????);
    free(??????);
    return 0;
}
```

a). n (It is n) (extra) * sizeof(int)
b). [illegible] (*d)[j] > (*d)[j+1]
c). int temp = (*d)[j]
d). (*d)[j] = (*d)[j+1]
e). (*d)[j+1] = temp
(12)
f). *b
g). **b

Q.4. Fill in the blanks for the following code. [6x2=12]

Output:
```
List: 20 -> 10 -> NULL
List: 10 -> NULL
```

```
#include<stdlib.h>
typedef struct dllnode * DLLNODE;
struct dllnode{
  int ele;
  DLLNODE next;
  DLLNODE prev;};
```

```
typedef struct doubly_linked_list * DLIST;
struct doubly_linked_list{
  int count;
  DLLNODE head;
};

DLIST createNewList(){
  DLIST myList;
  myList = (DLIST) malloc(sizeof(struct doubly_linked_list));
  myList->count=0;
  myList->head=NULL;
  return myList;
}
DLLNODE createNewNode(int value){
  DLLNODE myNode;
  myNode = (DLLNODE) malloc(sizeof(struct dllnode));
  ???????                                          a). _____myNode->ele= value;_____
  ???????                                          b). _____myNode->next = NULL;_____
  ???????                                          c). _____myNode->prev = NULL;_____
  return myNode;
}
void insertNodeIntoList(DLLNODE n1, DLIST l1){
  // case when list is empty
  if(l1->count == 0) {
    l1->head = n1;
    n1->next = NULL;
    n1->prev = NULL;
    l1->count++;
  }
  else {// case when list is non empty. Insert at beginning
    n1->next = l1->head;
    n1->prev = NULL;
    n1->next->prev = n1;
    ?????????                                      d). _____l1->head = n1_____
    l1->count++;
  }
}

void removeFirstNode(DLIST l1)
{
if(l1->count==0)
{printf("List is empty. Nothing to remove\n");}
else
{
DLLNODE temp=l1->head;
???????????                                        e). _____l1->head = temp->next_____
l1->head->prev = NULL;
free(temp);
l1->count--;
}
return;
}
void printList(DLIST l1) {
    if (l1->count == 0) {
        printf("List is empty\n");
    } else {
        DLLNODE temp = l1->head;
        printf("List: ");
        while (temp != NULL) {
            printf("%d -> ", temp->ele);
            ????????                               f). _____temp = temp->next_____
        }
        printf("NULL\n");
    }
}
int main()
{
DLIST newList = createNewList();
    // Create new nodes
    DLLNODE n1 = createNewNode(10);
    DLLNODE n2 = createNewNode(20);

    insertNodeIntoList(n1, newList);
    insertNodeIntoList(n2, newList);
    printList(newList);
    removeFirstNode(newList); // Remove first node and print list
    printList(newList);
    return 0;
}
```

---END OF QUESTION PAPER---